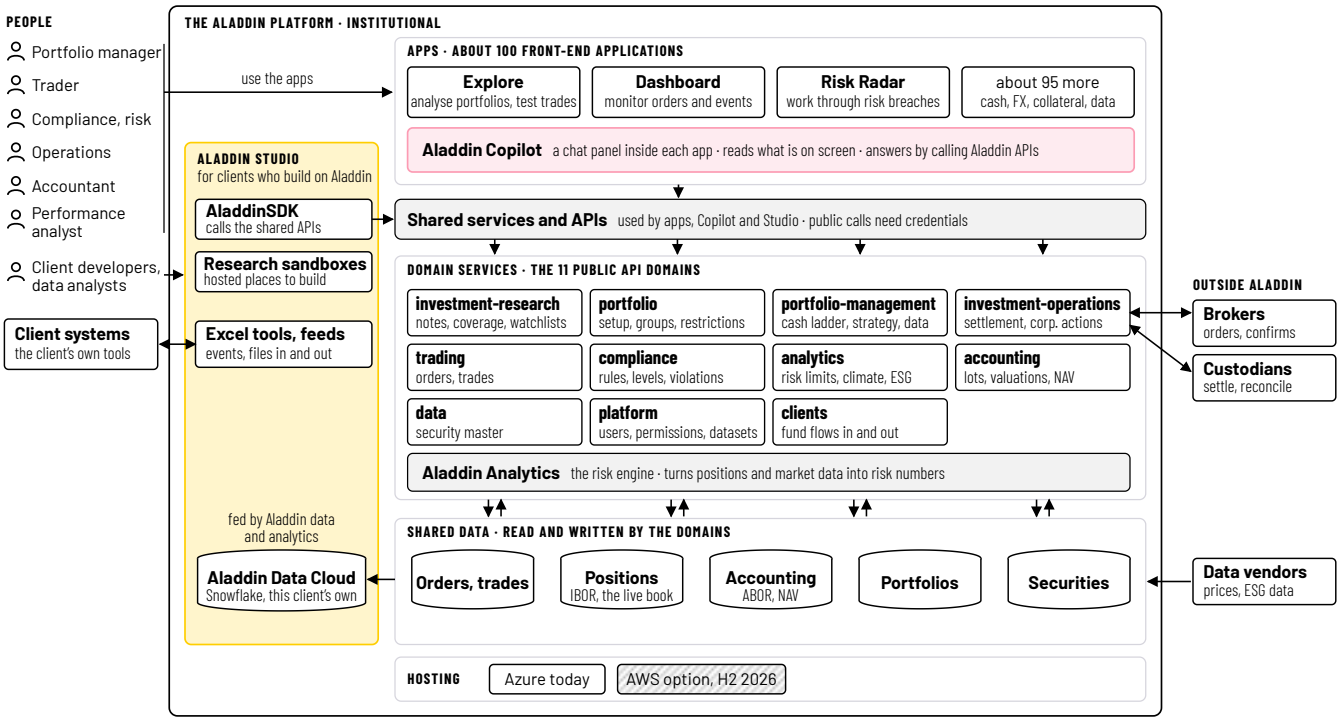
Exhibit 3. Apps, Copilot and Studio share the same Aladdin services and data



For illustrative purposes only. The 11 domains are the packages of BlackRock's public SDK, named as in its specifications. Other app names come from those specifications, so their screens are not drawn. ■ Source: Rosales and Vicente Valdez at 2:35, for the apps and Copilot[9]; Woodhead at 4:49, for Studio, events and the Data Cloud[1]; the public specifications and SDK[6, 29]; a training recording, for Explore[27]; BlackRock's Studio, APIs, ATX, ADX, Data Cloud and Aladdin Risk pages[14, 20, 30, 31, 32, 33]; the Data Cloud launch release[34]; hosting[21, 35]; data vendors, ICE and Sustainalytics[36, 37]. Solid outline = shown in sources; hatched = announced, not live.

Exhibit 4. Studio, grouped here into six pieces

Piece	What it is, in plain words
APIs (Aladdin Graph)	Ways for a program to read and write Aladdin data. Aladdin Graph[29, 38] is the whole API layer, not one API. Every public specification's id starts `agraph.<domain>`[6].
Event-driven workflows	Aladdin tells a program when something happens[31, 39]. Woodhead names "APIs events" as one part[1]; Diwakar Goel, Global Head of Data, Aladdin, wants "every single transaction … via an event, via an API"[40].
Aladdin Data Cloud	A managed data store on Snowflake, with Fivetran moving the data[14, 28]. Goel puts its latency at "within 15 minutes"[40].
Aladdin Compute	A hosted analytics platform with ready-to-use Python environments[41]. The SDK has a run mode for "Aladdin Studio Compute sessions"[29].
Aladdin SDK	An open-source Python library that handles login, retries, rate limits, pagination and long-running operations[29]. Eleven add-on packages, one per API domain, are generated from the specifications by BlackRock's plugin builder[42].
Developer Portal and Data Dictionary	The API catalog, framework guides, field dictionary, notebooks and code samples the APIs page describes[30]. A BlackRock job posting names the "Aladdin Studio Developer Portal"[39].

■ Source: BlackRock Studio and APIs pages[14, 30]; the 54 public specifications and SDK README[6, 29]; Woodhead[1], Goel[40] and Fivetran[28]. BlackRock gives no count. Its APIs page also links Aladdin Data eXchange and Aladdin Tools for Excel, which Exhibit 3 shows[30]. No event specification is public[6]. The APIs page describes the catalog, guides, notebooks and samples but does not link them[30]. The specifications link the guides to a portal behind a client login[6, 43].

Exhibit 3. Apps, Copilot and Studio share the same Aladdin services and data

Who builds	Client engineers	Code-curious portfolio managers	Data analysts	Excel users building reports
BlackRock's applications	**About 100 applications** such as Explore, Dashboard and Risk Radar	**Aladdin Copilot, inside them** each tool matches an Aladdin API already in production		
Aladdin Studio	**Aladdin Studio:** the developer and data layer; its parts are in Exhibit 4			
Shared services and data	**Portfolio management, trading, risk, compliance, operations, accounting** reached through the 11 public API domains			
Hosting	Microsoft Azure today	AWS option, announced		
Outside inputs	The client's own and third-party data, loaded into Data Cloud or through the Data eXchange, and vendor data such as prices and ESG research			

CONTROLS BUILT INTO THE APIS

Access set per API
Each API has its own access scope. A user sees only what their entitlements allow.

Compliance by default
The order API runs compliance by default and returns any violations.

A second approver
Dataset releases need a second check.

For illustrative purposes only. Every box is named in the sources below; the AWS option is announced, not live. The 11 domains are the packages of BlackRock's public SDK; the specifications also name Dashboard and Risk Radar. Controls as in Exhibit 14. ■ Source: Rosales and Vicente Valdez at 2:35, for the apps and Copilot[9]; a training recording, for Explore[27]; Woodhead at 4:49, for Studio[1]; the public specifications and SDK[6, 29]; BlackRock's Studio, APIs, ATX, ADX, Data Cloud and Aladdin Risk pages[14, 20, 30, 31, 32, 33]; the Data Cloud launch release[34]; hosting, Azure and the AWS release[21, 35]; data vendors, ICE and Sustainalytics[36, 37].

Exhibit 4. Studio, grouped here into six pieces, and the parts of a client's Studio named in the SDK files

WHAT STUDIO OFFERS

APIs (Aladdin Graph) Read and write Aladdin data. Every public specification's id starts `agraph.<domain>`	**Event-driven workflows** Aladdin tells a program when something happens. No event specification is public	**Aladdin Data Cloud** A managed data store on Snowflake, with Fivetran moving the data
Aladdin Compute Hosted, ready-to-use Python environments	**Aladdin SDK** Open-source Python library for login, retries and rate limits. BlackRock's plugin builder generates its 11 plugin packages from the specifications	**Developer Portal and Data Dictionary** Catalog, guides, field dictionary, notebooks and samples, behind a client login

The APIs page also links Aladdin Tools for Excel and the Aladdin Data eXchange

INSIDE A CLIENT'S STUDIO WORKSPACE, AS THE PUBLIC SDK FILES NAME IT

Projects Personal or shared. OAuth applications and storage are set up under one

S3 storage integration File storage set up in a Studio Project, its bucket assigned to a user or project

Studio notifications Email or push alerts to a user, a Project or a Space

ADC Admin Center Studio's screen for managing Data Cloud datasets

Studio Compute Sessions where SDK code can run, as an alternative to a developer's local machine

API tokens and API applications Sign-in details for programs: a token, or an OAuth client ID and secret

Studio UI Studio's screens, where tokens, applications and storage are set up

■ Source: BlackRock Studio and APIs pages[14, 30]; the 54 public specifications[6], for the ids, the portal link and the Admin Center; the SDK README[29], for the SDK and the lower group; Naik and Kalish, for Aladdin Graph[38]; the plugin builder[39]; Woodhead[1], Goel[40], the Aladdin Risk page[31] and a BlackRock job posting[41], for events and the portal; Fivetran[28]; Snowflake's partner page, for Compute[42]; the developer portal[43]. BlackRock gives no count of pieces. The APIs page describes the catalog, guides, notebooks and samples but does not link them; the specifications link the guides to a portal behind a client login.